

Pre-commercial Procurement (PCP) FABU LOS

Future Autonomous Bus Urban Level Operation System

TENDER DOCUMENT 9: Field Test Specifications

These Field Test Specifications, designated as Tender Document 9, should be read in conjunction with other documents related to this Pre-Commercial Procurement (PCP), listed hereunder:

- Tender Document 1: Request for Tenders
- Tender Document 2": Functional Specifications
- Tender Document 3: The Framework Agreement, signed by all Suppliers on 11 December 2018
- Tender Document 7: call-off for Phase 3 - Field Testing
- Tender Document 8: The Specific Contract for Phase 3
- Forms A", E", E" bis, F" and G"

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1. Overview

Phase 3 will have a maximum duration of 9 months. The Procuring Partners intend to sign contracts with up to 3 Suppliers. During that period, each Supplier will conduct two pilots. The duration of one pilot is minimum 50 operational days per pilot site. The pilots are intended to run (at least) 6 hours a day. The FABULOS Procuring Partners reserve the right to adjust the time schedule if necessary. This will be communicated in a timely manner to all Suppliers.

In Phase 3, field testing of the prototypes takes place on predetermined pilot routes in the Procuring Partner cities. Each pilot will have an interim evaluation and payment as well as a final evaluation and payment. The output is the successful completion of two field tests per Supplier. These field tests will be done consecutively. The aim is to have stable field tests that work on a technical level and have been tested in real-life settings with end-users, described in an End-of-Phase Report and demonstrated to the Procuring Partners as input towards a potential future procurement procedure after FABULOS has run its course.

1.1 The pilot cities

The specific set-up and circumstances of each field test will be further detailed in this document. You can find information about local regulations and exemption process in each Procuring Partner city for automated vehicles, together with information about each city's open API's, information on wireless network capabilities, on the vendors and technologies currently used for traffic lights and on the availability of route-wide remote camera surveillance.

Field testing in Phase 3 needs to be done in 2 cities. There is budget for 3 Suppliers to conduct field tests in Phase 3. At a minimum, there should be 2 Suppliers. Due to the (inflexible) PCP process, it will only be known in January 2020 if there will be 2 or 3 Suppliers proceeding to the Field Testing Phase.

Now that Porto is no longer a field testing city, there are 5 cities left: Gjesdal, Helsinki, Lamia, Helmond and Tallinn. The first group of pilot cities - having field tests in Spring 2020 - are Gjesdal, Helsinki and Tallinn.

The national legislation and pilot permissions processes in each country will have a crucial impact on the possibility to implement the pilots.

1.2 Overview of the two pilot rounds

Round 1	Timing	Activity
Gjesdal	Between 1.4.-30.6.2020*	Field Tests according to the specifications in Tender Document 2''
Helsinki	Between 1.4.-30.6.2020*	
Tallinn	Between 1.4.-30.6.2020*	

**minimum 50 operational days. There is some flexibility regarding the start and end date of the pilot (can start before April and end after June, provided this does not compromise the second pilot)*

Round 2	Timing	Activity
Helmond	Before 31.10.2020*	Field Tests according to the specifications in Tender Document 2''
Lamia	Before 31.10.2020*	
Helsinki	Before 31.10.2020*	

**minimum 50 operational days. There is some flexibility regarding the start date of the pilot, but the end date of 31.10.2020 is a hard deadline and cannot be changed.*

1.3 Step-by-step Field Test planning

The overview below is a more elaborate version of the key steps per stage. This is a tool to help consortia with the planning of their Field Tests; it will also be used for future reporting on Function 9 (Tender Document 2'', Functional Specifications; Function 9: Deployment, set-up and service) and some other Functions.

A. Obtaining permissions

As per earlier guidance and city webinars, Suppliers are to obtain all necessary permissions, insurances and test-plates as soon as they are notified of their two pilot locations (relates to all subfunctions of **Non-functional Requirement 3**).

B. Pilot preparation phase

The following step-by-step guidance starts from the assumption that there is permission to operate an automated vehicle in road traffic. That is to say, the necessary test plates and related insurance have been acquired as well as any other relevant permissions.

1. Planning the route

- To be done in cooperation with your site-specific local contact person from FABULOS
- Planning the route (the route itself, bus stops, storage and charging place, etc.).
- Presenting the route to the bus supplier (route map, pictures, videos etc.).

- Supplier visit on site, if needed.
- Supplier makes site analysis of the route where the requirements that have to be materialized are listed in order for the route to be implemented (these can be for instance additional infrastructure or traffic signs to intersections).
- Finding a place for RTK GNSS-base, if needed (usually a roof of the highest building in the area). Recommended option is to use a local virtual reference station (VRS) subscription provider. No need for a fixed RTK base as the satellite corrections come through mobile network. This technique has been used in pilots in Helsinki in the past.

2. Discussing the route with the city or landowner

- For instance, discuss possibility to remove parking spaces for autonomous shuttle bus stops and making additional arrangements.
- To be done in cooperation with your site-specific local contact person from FABULOS

3. Discussing the route with local PT authority

- To be done in cooperation with your site-specific local contact person from FABULOS
- If there are other PT vehicles driving on the same path, PT authorities usually hope that the robot bus(es) does not disturb the traffic.
- Planning the schedule of the robot bus for synchronising it with other PT vehicles (schedule of the robot bus cannot necessarily be known before the route has been programmed and tested as there are many things that affect the time it takes to drive the route through).

5. Informing local transport safety agency

- To be done in cooperation with your site-specific local contact person from FABULOS
- Only in case this institution was not yet involved in the permission process.

6. Planning all the necessary traffic arrangements for the route

- Traffic arrangements plan for the traffic signs and additional infrastructure. A local traffic arrangement company can be consulted on this.
- Agreement with the city or landowner
- To be done in cooperation with your site-specific local contact person from FABULOS

7. Making all the necessary traffic arrangements for the route

- Traffic signs and other required infrastructure. Your site-specific local contact person from FABULOS can help and a local traffic arrangement company can also be consulted on this.
- Specific bus stop signs for the robot buses or leaflets to normal PT bus stops, if these are used as bus stops (schedule can be difficult to know before the actual operation)
- Make sure it is clear if/how traffic lights communicate with the buses.

8. Making sure that the storage/charging place is ready for the buses

- Site-specific local contact person from FABULOS will bring consortia into contact with the owner/contact person of the storage/charging location.
- Height of space: make sure buses fit.
- Check that there is sufficient space for 3 buses at the same time.
- Safe and secure: make sure it is clear how to gain access and who has access.
- Check warm / cold water for cleaning
- Check that there are suitable charging facilities

9. Acquire unlimited data subscriptions (SIM cards) for the bus and RTK-base

- The responsibility of the Suppliers, but assistance may be needed in foreign countries. In that case, be in touch with the local FABU LOS contact person for that particular site.
- In Finland, it was Metropolia University of Applied Science that has acquired these in previous pilots.

10. Transport of vehicles to pilot site

- Obtain transportation quotations if needed
- Take potentially long delivery times into account
- Someone to receive the vehicles and show storage/charging place and the route.
- The Lead Procuring Partner (FV-Helsinki) has a small budget to cover (some of) the transportation costs of the vehicles to the pilot site. To discuss this before paying a transportation company.

11. Fixing the RTK-base

- In case of a RTK-base, the sSupplier needs access to the building's roof and must have permission to install the equipment. Further access during the pilot may be needed as well.
- If VRS technique is used, there has to be a subscription available to upload VRS data in the bus. These are probably also tested beforehand together with the bus supplier and VRS subscription provider.

12. Mapping the route

- Mapping the route with the bus itself or a different type of system.
- If there is busy traffic on the route, the mapping may need to be done during nighttime, as other vehicles may disturb the map quality.
- Mapping can usually be done during one day.

13. Programming of the route

- Route is driven through several times and points such as bus stops, intersections and zebra crossings are programmed.
- Bus stops are to be agreed upon with your site-specific local contact person from FABU LOS
- Speed is increased step by step.
- Programming can take several days or weeks depending on the route complexity.

14. Setting up control room

- Space is provided by the FABU LOS Procuring Partners (desks, chairs, electricity, wifi, toilets) near the pilot route. For changes or wishes, contact the site-specific local contact person from FABU LOS.
- Suppliers are allowed to *not* make use of this provided control room and use one of their own. The control room can be located by the operational area but it is allowed to locate it off-site as well.
- Install all necessary equipment (screens etc).
- If applicable, make sure there is space for a local incident team - in the control room or elsewhere

15. Integration with local PT route planners

- In Helsinki this has been done by informing Helsinki Region Transport (HSL, the public transport operator) about the route and informing about the schedule. HSL has then uploaded the timetables to the local Public Transport Journey Planner.
- To be done in cooperation with your site-specific local contact person from FABU LOS, who will bring Suppliers into contact with relevant local Public Transport Operator staff.

- The open HFP API can be used to subscribe to vehicle movements in soft real time so that the buses can be seen in real time in a Journey Planner (<https://digitransit.fi/en/developers/apis/4-realtime-api/vehicle-positions/>). .

C. Implementation phase

1. Monitoring

- Passengers (establish way of counting them)
- Mileage
- Operational hours driven (hrs.)
- Amount of automated and manual driving (%)
- Energy consumption (during the drive, charging energy, specify the consumption of air conditioning/heating?)
- Amount of operator intervention and the situations specified
- Amount of emergency stops
- Realization of the planned timetable (departures, days, hours). What is considered an undriven departure?
 - Example: According to Helsinki's local Public Transport Operator, a departure is considered undriven when a departure has not started before the next departure operating on the same direction, or within 15-20 min of the planned departure.
- Use of the on-demand feature (if available)
- Average speed

2. Reporting

- Report on the points mentioned in C.1 above (as well as on general pilot progress and challenges) - in the least during mid-term pilot evaluations and end-of-pilot evaluations (with reference to the Deliverables described in Tender Document 7). Templates will be provided by the Procuring Partners in due time.
- We foresee weekly "update" emails between the Supplier and the site-specific local contact person from FABULOS during the preparation stage and during the actual piloting
- We foresee bi-weekly "update" telephone calls between the Supplier and the site-specific local contact person from FABULOS during the preparation stage and during the actual piloting

3. Communication activities

- In cooperation with the FABULOS site-specific local contact person.
- Take guidelines from the FABULOS Communications team into account.
- Agree on division of tasks between Consortium members and FABULOS team (e.g. related to media contacts; "marketing" videos of pilots; presentations in conferences, materials) and planning.
- To report all communication activities to the site-specific local contact person and the FABULOS Communications team (template will be provided).

4. Customer surveys

- Part of Non-Functional Requirement 2.1 (TD2"). The Supplier is expected to carry out at least one user satisfaction / acceptance survey among a predetermined minimum number of passengers during the Field Testing. Procuring Partners will carry out before and after surveys in their respective cities.
- All surveys will consist of several mandatory questions (equal for all pilot sites) plus the option to add own questions.

- Engage disabled people as part of the survey group.
- Determine who will do the surveys and how. E.g. a person inside the bus or an online link shared for the passengers, QR code, something else?
- Seek cooperation with local/national traveller organisations and/or organisations for disabled people (with reference to NFR2.5) to e.g. determine suitable survey questions, engage/inform their members, invite representatives for test-drives and in general get feedback on your solution.
- Seek cooperation with a University of other stakeholder (with reference to NFR2.5) to analyse and report on the survey results

5. Crisis communications

- Refers to NFR1.10 related to Crisis Communication Plan
- To be done in cooperation with your site-specific local contact person from FABULOS
- Send all relevant actors the agreed-upon procedure in case of an accident.
- Inform all people involved in the crisis communication chain about the exact start date and route of the pilot.

6. Deviations in the operating timetable

- In case of a fixed time schedule: deviations are seen via the local PT journey planner
 - Example: Helsinki's public transport operator HSL has (during previous pilots) not been very happy with receiving multiple deviation notifications. In case of longer technical failures, the routes have been completely removed from their systems and reloaded again when it is certain that the buses can operate again.
- Consider putting up an own channel for the pilot, where deviations can be reported (example: <https://www.helsinkiroboline.fi/poikkeusinfo/>) . To be done in cooperation with your site-specific local contact person from FABULOS
- Determine who will make the deviation notifications.
 - Example: in HSL traffic the operator companies have access to an Operational Monitoring and Management system (OMM) which is used through a browser. These deviations are then automatically transferred to the Journey Planner as the deviation has been registered through OMM.

7. On-demand

- Refers to function 1.9 (TD2''), which is a nice-to-have function: "Routing on the fly to go beyond fixed routes (within pre-specified area of operation), according to passenger on-demand requests".
- Also refers to function 8.10 (TD2''), which is a nice-to-have function: "Can be operated on fixed routes, stopping anywhere on the route by on-demand request from inside or outside the bus (e.g. mobile app) with safety considered"
- The operational area of the on-demand service needs to be mapped out and taken into account well beforehand in the deployment planning
- Check if the on-demand feature can be integrated in the local public transport route planner
- Communicate the availability of this option to (potential) users
- Bullets above to be done in cooperation with your site-specific local contact person from FABULOS

2. City Introductions

2.1 Gjesdal

Gjesdal is a mountainous district, on the west coast of Norway, approximately 30 km southeast of Stavanger. The population of Gjesdal has passed 12 000, and is spread over an area of 640 km². Ålgård is the county's town centre, with 10000 inhabitants.

Gjesdal strives to put its inhabitants first, whilst also developing commercial activity. We aim to provide a holistic service for the user, where internal departments collaborate to provide best service and with the most effective use of resources. Our vision is "We go further" and our main objective is to ensure that Gjesdal is a great place to live, work and do business.

The main road E39, passes through Ålgård. E39 leads to Sandnes and Stavanger in the north and to Kristiansand in south. The road is used by thousands of motorists and lorries every day, as it is the main connection between the east coast and west coast of Norway.

2.1.1. Permissions and exemptions for pilots, Gjesdal

Describe legislative matters incl. required licenses and local permission processes.

Ecosystem: overview of main Stakeholders - who decides what?

Who needs to be notified? (PTA / PTO (API's, routeplanners), road safety institute, emergency services?)

[Law](#) and [regulations](#) from 1.1.2018.

Application form still in progress (temporary form – checklist) and only in Norwegian at the time.

Application and attachments to be send to the [Norwegian Public Roads Administration](#) postmottak@vegvesen.no with marked «Søknad om utprøving av selvkjørende motorvogn».

Application on English is accepted.

The only public transport system from/to Gjesdal is bus lines operated by Kolumbus.

On this [link](#) you can see in the real time map were the bus is right now (bus line 23 goes from Sandnes to Ålgård)

Vehicle class / types of registration plates needed?
 Operation allowed with passenger car driving license?
 Insurance issues? (national business ID needed to obtain suitable insurances?)
 Is remote operating allowed? For more than 1 vehicle?
 Liability: operator? company? software owner?
 Mandatory seat belts for field tests? Standing places allowed?
 Ongoing legislation that could have an effect on the field testing?

Vehicle class as for other vehicle.

- M1 Passenger car (Maximum 9 seats)
- M2 Bus (Maximum total weight 5,000 kg, at least 10 seats)
- M3 Bus (Permitted total weight over 5,000 kg, at least 10 seats)

Ordinary registration plates.

Foreign company can import the vehicle, no tax on electric vehicle at the time.

[Car approval and car trailer regulations](#)

Application timeline estimate - explain steps to take
 Application submission: where, how? In English?
 Application content / field test plan / Safety and security plan (content; done by whom?)

Application must include

Information about applicant

- Security officer/manager
- Operator
- Information about vehicle
- Insurance
- Time and place for the pilot
- Information if extra signs, markings, signals, railings/fence etc. is needed
- Any need for exemption from law and regulations in "[vegtrafikkloven](#) med [forskrifter](#) and [yrkestransportloven](#) med [forskrifter](#)":
- The purpose of the pilot.

Attachments

1. Plan for trial
Detailed description of activities with times, plan / map outline of the relevant area (industrial area / airport / pedestrian cycle path / motorway / traffic lane / traffic lights / highway / crossings / speed limits / road users involved...)
2. Insurance documents: insurance certificate and agreement.
 Same as other countries, non Norwegian company need to get a [Business ID for Foreign company](#) called NUF
3. Card of customs declarations if the vehicle shall be registered in Norway
4. Copy of the Vehicle Card if the vehicle has been registered in Norway or another EU country. If European type approval, the type approval number.
5. Documentation of all/or some the technical terms above.
6. Possibility to apply for exemption from some of them

7. *Function Description*
8. *System Description*
9. *Technology description*
10. *Description of the maturity of the system*
11. *Description of how information security and privacy are taken care of.*
12. *Description of the risk of the automation functions*
13. *Explanation of the system's electromagnetic compatibility (EMC).*
14. *Risk assessment - The Roads Authority may require the report to be approved by an independent third party.*
15. *Requirements for operators - Documentation: Certificate + certificate from the manufacturer that the operator has adequate training. Describe the operator's role regard to the driver's responsibility and the relationship with the safety officer.*
16. *Other relevant information*

2.1.2 Pilot route and implementation, Gjesdal

- Rationale for this route
- Characteristics (incl technical level of traffic light systems, difficult spots & possible infrastructural changes)
- “Plan B route” with lower ambition level should be available as back-up)

No traffic lights. Large incline and two/three roundabouts. Part of route with speed limit 50 km/h. Possible to shorten the existing route and make it easier due to incline and length.

- Has been discussed and confirmed with (authorities & stakeholders x, y, z)
- Support with legislative matters incl. required licenses and local permission processes.
- Information on city’s software system APIs (depending on technical maturity of each city)

The route is proposed and confirmed by the municipality of Gjesdal. For all related matters, there is a single point of contact to Gjesdal municipality -Trond Hansen – Trond.hansen@gjesdal.kommune. Other important stakeholders are:

- Road authority, who are aware of the project : Statens vegvesen, Trafikketaten
- Local PT operator Kolumbus

- Storage & charging facilities for buses (Storage facilities for the fleet for off-duty hours, electricity for charging (standard socket, 230 V) and s
- Control room location / facilities, pace for control room close to the pilot zone for 2-3 persons.
- Mitigation measures city can/should take; supplier can/should take, Support of local authority’s transport planners for the plan on the pilot area & routes.

Charging point will be close to the city center. Storage facility and control room location will be in the city center 10 m from the test route and with view to a part of it. A two floor building will host storage and control room functions.

- Please describe the city’s open API’s (only the ones that are directly related to mobility/ real time location etc)
- Please describe the city’s information on wireless network capabilities related to piloting
- Please describe the city’s availability of route-wide remote camera surveillance
- Please describe the city’s on the vendors and technologies currently used for traffic lights.

Kolumbus the PT has some open data on [this link](#). Gjesdal has [some open data](#), but not on public transport at the moment.

There is a wireless network in the city center.

No camera.

No Traffic lights on the route or in the city.

2.2 Helmond

The City of Helmond is a medium-sized city (90.500 inhabitants) in the southeast of the Netherlands. It is part of the Brainport region which is one of the three most important economic areas in the country. Helmond is home to the Automotive Campus, where over 35 companies work on innovative mobility solutions. With the Automotive Campus, Helmond presents itself as knowledge and innovation centre for mobility solutions and as "City of Smart Mobility". Though a relatively small city, Helmond receives international recognition for its ambitious goals and the realisation of innovative mobility solutions in the city. A recent example of this is winning the local government award during the ITS World Congress 2019 in Singapore.

The link below gives a good impression of our activities:

<https://www.youtube.com/watch?v=AczObRU6Y9Y&feature=youtu.be>

With all developments in the field of increasing traffic, climate, safety and liveability problems, the need for alternative and sustainable means of transport (accessible for everyone) is more urgent than ever. With 4 train stations, Helmond is well connected to national and international destinations. The challenge in Helmond is mainly on the last mile connection. Approximately 2/3 of the city lies within the last mile distance from a train station.

2.2.1. Permissions and exemptions for pilots, Helmond

Describe legislative matters incl. required licenses and local permission processes.

Ecosystem: overview of main Stakeholders - who decides what?

Who needs to be notified? (PTA / PTO (API's, routeplanners), road safety institute, emergency services?)

There are two ways two apply for a permit.

- 1) *Applying via the BOEV procedure The BOEV procedure is already longer available and was updated in 2015 to make it possible to test with self-driving vehicles, but only with a driver onboard.*
- 2) *Applying via the new experimental law As of the first July 2019 it's possible to apply for a permit where you can test with self-driving vehicles without a driver / steward on board and thus only with a remote driver.*

Taking into account all aspects, we believe the consortium should apply via the BOEV procedure in Helmond. This means that there is also a steward / driver on board. In the case of the BOEV procedure, the RDW (the National road authority in the Netherlands) is issuing the exemption.

The permission process and the main stakeholders in this process are described in the paragraph with "application timeline estimate - explain steps to take". Other organisations that needs to be notified are the public transport authority (province of North Brabant), the public transport operator (Hermes) and the emergency services. The city of Helmond keeps those organisations informed.

Vehicle class / types of registration plates needed?
 Operation allowed with passenger car driving license?
 Insurance issues? (national business ID needed to obtain suitable insurances?)
 Is remote operating allowed? For more than 1 vehicle?
 Liability: operator? company? software owner?
 Mandatory seat belts for field tests? Standing places allowed?
 Ongoing legislation that could have an effect on the field testing?

Vehicle class / types of registration plates needed?

The Category depends on vehicle:

Category M1: Vehicles designed and constructed for the carriage of passengers and comprising no more than eight seats in addition to the driver's seat.

Category M2: Vehicles designed and constructed for the carriage of passengers, comprising more than eight seats in addition to the driver's seat, and having a maximum mass not exceeding 5 tonnes.

Category M3: Vehicles designed and constructed for the carriage of passengers, comprising more than eight seats in addition to the driver's seat, and having a maximum mass exceeding 5 tonnes.

Experimental vehicles that have a special permit from the RDW start the license plate with ZZ.

Existing vehicles with a number of innovative modifications can have their current number plate and have an exemption for the innovative components. Please note that the existing license plate is valid in the Netherlands.

Operation allowed with passenger car driving license?

Yes but not preferred. There is no strict regulation for this. However, in these projects it is all about risk management. So an experienced person, also in driving with buses, is of course a plus.

Insurance issues? (national business ID needed to obtain suitable insurances?)

During the preparation phase the applicant must demonstrate that the vehicle is adequately insured. The test applicant proposes an insurance company offering a guarantee for the insurance of the test.

Is remote operating allowed? For more than 1 vehicle?

Yes, but in Helmond we go for a Field test with driver / steward onboard.

Liability: operator? company? software owner?

As part of the "experimental law" (which makes it possible to drive with remote drivers) all obligations and responsibilities that apply to drivers in the vehicles will also apply for drivers outside the vehicle. The "driver" is thus liable for driving the vehicle. He or she is primarily responsible, but other parties can also be held responsible if things are not in order in the preparatory process or otherwise. In the case of the BOEV procedure this is probably the same. The driver is primarily responsible, but every organization involved in the process can also be held responsible for their role in the process.

Mandatory seat belts for field tests? Standing places allowed?

- *For safety reasons please use seat belts, of course it's important that these also meet the quality requirements.*
- *Regarding standing places: With certain conditions the answer is yes, but the bus needs facilities to keep the person stable (like sturdy tubes). It is also very important for these vehicles that people with a wheelchair have steady places in the bus (and please take into account driving in two directions). For the field test it's for safety and mitigating measures better to only use seats.*

Ongoing legislation that could have an effect on the field testing?

Self-driving vehicles are relatively new and in development, so the answer is Yes: legislation or procedures for permits can change.

Application timeline estimate - explain steps to take

Application submission: where, how? In English?

Application content / field test plan / Safety and security plan (content; done by whom?)

The applicant must request an exemption from the RDW. You can contact directly our contact person:

Mr. Pieter van der Stoep, account manager Automated and connected Driving

M +31 646384461

E pvanderstoep@rdw.nl

The application and procedure can be done in English.

The whole exemption process can be divided into five steps under the direction of the RDW:

- 1) **The intake** step is focused on the preparatory treatment and pre-assessment of the test application;*
- 2) **The preparation** step is focused on the preparation of the assessment of the test application;*
- 3) **The assessment** step provides a decision on safety and whether the test can and may be carried out;*
- 4) **The execution** step is dedicated to carrying out the test on public roads;*
- 5) During the evaluation step, it is reviewed to see whether the process must be improved and to safeguard knowledge.*

As a separate document we already shared with you the Assessment framework from the RDW. In this document the entire process is described in more detail.

Intake

For the intake the applicant is expected to submit an information document based on the intended test. This document contains the following:

- A fully worked out plan of action;*
- A fully completed application document;*
- Declaration to cooperate in the research.*

The RDW can request additional information.

Test applications must be submitted via the contact person of RDW. There will be an intake interview between the RDW and the test applicant. This meeting will cover the process, expectations and confidentiality. Every test application will be allocated a contact person within RDW who is both the point of

contact for the applicant and coordinates the process with the other primary, secondary parties and the applicant. Agreements are communicated by the contact person.

Preparation

The test applicant himself analyses the risks and the mitigating measures. This can be done with the aid of qualitative documentation such as ISO 26262 and Failure Mode and Effects Analysis (FMEA analysis). Any other documentation is permitted in consultation with the RDW. In the analyzing the application divides the risk and mitigating measures in 3 categories: Vehicle, Behaviour and Other things.

Driving on public roads with CAD entails risks, where the innovative factor can have a (positive and/or negative) influence on safety. To anticipate this, the reasoning is based on the Road Traffic Act 1994 and the RVV (this is Dutch traffic laws). Legislation and the RVV include codes of conduct which must be adhered to in order to be able to participate safely in traffic. The test applicant shall demonstrate that it can comply with this. This can happen in different ways:

- Published vehicle requirements
- Exemption of requirements
- Vehicle with a foreign exemption
- Alternative solutions

The first one means that maybe several elements of the vehicle already have type approval or can pass an individual vehicle test. The second one means that not all components need to be important in the context of testing. For example, if you only drive during the day, then lighting is perhaps of less importance. The third one means: A previously granted licence in other countries can be used to demonstrate safety. This on the condition of applicability to the application document of the test application and clear and qualitative documentation. And the 4th one means: If the test applicant can demonstrate that the same safety level (or higher) is realized with the published vehicle requirements, it can receive an exemption for the relevant objective. These alternatives can be prepared based on the vehicle, people and the environment.

Kick-off meeting

As soon as it is clear which secondary parties are required, a kick-off meeting is held. This meeting is organized by the RDW, whereby the RDW determines and approaches the parties. The primary and secondary parties and the applicant must be present. In this meeting, all information is shared and where necessary explained. Confidentiality of documentation is discussed. Agreements made must be clear and communicated with all stakeholders. In follow-up to this agreement, knowledge questions are prepared appropriate to the application document of the test.

Involved originations in RDW-process:

- The RDW assesses the technology used in the vehicle.
- The SWOV (this is the institute for road safety research) advises the RDW about the behavior component
- The local road authority (city of Helmond) grants substantiated permission for the use of the road including from a road safety and traffic flow point of view. The road authority can make use of the Taskforce Dutch Roads (This is a partnership consisting of various organizations that help road authorities by developing knowledge and sharing experiences of tests with self-driving vehicles).

- Any other advisory parties, they can be requested to advise on aspects which cannot be assessed by the RDW, SWOV and the road authority. These parties could be focused on specific aspects, ElectroMagnetic Compatibility⁸ (EMC) for example, or the integral safety of the vehicle, human, road combination.

To complete the preparation phase you need: the Information document (from step 1: the intake), a qualitative safety documentation; Guarantee of insurance; and formulated knowledge questions during the kick of meeting.

Assessment

As mentioned as part of the preparation phase the stakeholders will each assess a part of the process. They all give advice to the RDW and they decide how to use the advice. Beyond coordinating the procedure the RDW will also test the vehicle and will carry out a physical test. All relevant stakeholder are present. During this physical test, two scenarios are worked through which are as close as possible to the test.

- *Happy test:* The carrying out and assessment of the vehicle at a closed site under ideal conditions for the performance of the test and the functioning of the described risk analysis of the automated functions.
- *Stress Test:* The carrying out and assessment of the vehicle at a closed site under the least ideal conditions for the performance of the test and the functioning of the described risk analysis of the automated functions.

To get an approved vehicle the RDW follows a process of 5 subjects:

- Part A: 'the characterisation tests' also known as 'nominal' or 'Happy flow tests' and
- Part B: single failure stresstests and
- Part C: the compliance with Annex 4 of European Type Approval Framework Directive 2007/46/EC and
- Part D: Compliance with ISO 26262 and
- Part E: Compliance with the Vehicle Safety and Security Framework*

RDW and the manufacturer work together in this process.

* The Vehicle Safety and Security Framework: New vehicles have more and more advanced automated functions. The RDW has developed the Vehicle Safety and Security Framework to assess a vehicle that is full of software. The framework is a collection of points for attention to check whether the manufacturer's processes in terms of software quality are sufficiently in order. The framework covers the entire life cycle of a vehicle and focuses on safety, functionality, security and privacy. The document contains approx. 500 questions about various aspects, from design to use. The questions help the RDW to determine to what extent the technology is reliable and is safe. The RDW works together with manufacturers, governments and knowledge institutes to develop and test this framework.

Execution

- The applicant provides weekly feedback through a logbook
- Adjustments to the vehicle during the exemption are permitted in consultation with the RDW.

- *In the event of a collision with (minor) material damage or (minor) personal injury, in or outside the vehicle, driving with the exemption is suspended with immediate effect. This is communicated to the RDW. The use of the exemption may be resumed after consent of the RDW and the road authority.*

Evaluation

During this phase: Evaluation of the process, testing and answering of knowledge questions as prepared in advance.

Duration of the application procedure:

The application processing time depends on the type of vehicle, completeness and quality of the documents submitted by the applicant, experience from earlier tests, the cooperation of the relevant applicant and the primary and secondary parties, as well as the observance of the handling time. The procedure can be done in 6 months, and when everything is going smooth faster. A big risk is using vehicles without sufficient type approvals. This can cause serious delays.

Support

The municipality of Helmond understands that the procedure for obtaining the required exemption in the Netherlands is extensive and robust and can therefore be time-consuming and labor-intensive. That is why the municipality of Helmond intends to engage external expertise that can assist the consortium on behalf of Helmond in the RDW procedure.

2.2.2. Pilot route and implementation, Helmond

- Rationale for this route
- Characteristics (incl technical level of traffic light systems, difficult spots & possible infrastructural changes)
- *“Plan B route” with lower ambition level should be available as back-up)*

“Automotive and Smart Mobility” is one of the strategic pillars of the municipality of Helmond. It is therefore no coincidence that the Automotive Campus is located in Helmond. At this campus companies, startups, knowledge institutes and governments work together to create the future’s mobility. A good accessibility for the campus is a must and needs to be improved.

Part of this accessibility is the connection with public transport. The city has 4 train stations but we need a sustainable last mile solution for commuters traveling by train. We believe that autonomous shuttles are the perfect solution for this need. And of course the Automotive Campus is a perfect location to demonstrate the possibilities of this kind of innovative solutions.

Pilot route characteristics:

- Distance 2.9 km
- Driving time single route, approx. 10 min.
- 2 roundabouts, no traffic lights
- Light slopes regarding overpass
- Maximum speed limit is 50 km/h over a distance of 500m. As mitigating measure we can lower the speed limit to e.g. 30 km/h on this stretch of road and make it also impossible for other road users to overtake. On the other parts of the route lower speeds are sufficient.
- Route is in mixed traffic with motorized vehicles (also some agricultural vehicles), cycles, pedestrian.
- 1 stop at Brandevoort station and 1 or maybe 2 at Automotive Campus

Difficult parts on the route is something also to be determined by the consortium. Helmond believes the list below is an overview of the most difficult locations / situations:

- *Passing a secondary school with on specific times lots of students on the road (most of them with a cycle)*
- *Passing a roundabout (are extra sensors needed?)*
- *Interaction with cyclists and pedestrians on the route, even an approaching agriculture vehicle is possible.*
- *Parked cars next to the road (as mitigation measures we can set parking bans during the field test)*
- *The situation on some intersections with motorized vehicles and/or with cyclist and pedestrians.*
- *Stretch of road (500m) with current speed limit of 50 km/h and many drivers are speeding here. (We need mitigating measure.)*

- *In the current situation there is a infrastructural measure to prevent cars from driving via the viaduct. We need to change this measure to make it possible for driving with the autonomous minibuses.*
- Has been discussed and confirmed with (authorities & stakeholders x, y, z)
 - Support with legislative matters incl. required licenses and local permission processes.
 - Information on city's software system APIs (depending on technical maturity of each city)
- *Involved are PT authority (province of North Brabant), PT operator (Hermes), Fire department (VRZBO, on behalf of emergency services), National Road Authority (RDW), Automotive Campus, Reizigersoverleg Brabant (interest group that represents travelers), SmartwayZ.nl (regional partnership focused on smart mobility).*
 - *As extra support the municipality of Helmond is exploring the possibilities of engaging external expertise that can assist the consortium in the Dutch exemption (RDW-)procedure.*
- Storage & charging facilities for buses (Storage facilities for the fleet for off-duty hours, electricity for charging (standard socket, 230 V) and s
 - Control room location / facilities, pace for control room close to the pilot zone for 2-3 persons.
 - Mitigation measures city can/should take; supplier can/should take, Support of local authority's transport planners for the plan on the pilot area & routes.
- *Storage & charging facilities for buses will be available on the Automotive Campus site. The city likes to hear from the consortium whether additional facilities are desirable. We will try to provide this.*
 - *Control room is also available on the Automotive Campus premises.*
 - *For every aspect that involves risks it's good to estimate the chance and impact and explore possible mitigating measures. Also for the exemption process in the Netherlands we need to reduce the risks as much as possible and thus work out mitigating measures. Especially the consortium and the municipality have to work together in this. For extra support the municipality is exploring the possibilities of engaging external expertise that can assist the consortium in the Dutch exemption (BOEV-)procedure. This possible assignment also includes support for a risk assesment and mitigation measures for the pilot route.*
- Please describe the city's open API's (only the ones that are directly related to mobility/ real time location etc)
 - Please describe the city's information on wireless network capabilities related to piloting
 - Please describe the city's availability of route-wide remote camera surveillance
 - Please describe the city's on the vendors and technologies currently used for traffic lights.
- *API's public transport in the Netherlands: Within NDOV (National Database Public Transport), transporters, public transport authorities and ticket offices work together to make source data from / about public transport available. For more information, email: meldpunt@crow-ndov.nl (API)*
 - *Helmond has many smart traffic lights (with ITS-G5) in the city, but there are no traffic lights on the Field Test route. Therefore not applicable.*
 - *4G coverage*

2.3 Helsinki

Helsinki is the Capital of Finland and the centre of the Helsinki Region with 650 000 inhabitants (1,5 million in the metropolitan area).

Helsinki City Strategy is to be “The most functional city in the world”. Helsinki also aims to become a carbon neutral city by 2035. Achieving this goal will require the City to conduct a variety of trials in order to find the best emission-free solutions. In the process of doing so, the City also aims to create the most effective transport services in the world. For residents, this will mean carbon neutral transport options and easy-to-use services, while companies can look forward to a world-class test environment for smart traffic. These development efforts will also provide the City of Helsinki with a competitive advantage and the science community with a fruitful research environment. The City of Helsinki has also worked systematically since 2009 to facilitate access to public information for use by the private sector. There have already been several robot bus trials conducted in Helsinki.

2.3.1. Permissions and exemptions for pilots, Helsinki

Describe legislative matters incl. required licenses and local permission processes.

Ecosystem: overview of main Stakeholders - who decides what?

Who needs to be notified? (PTA / PTO (API's, routeplanners), road safety institute, emergency services?)

Ecosystem

- *For all piloting or route related questions there is a single contact point towards the city → Ulla Tikkanen, ulla.tikkanen@forumvirium.fi, +358 41 526 1617.*
- *The Helsinki city as the land owner has agreed to host the pilot, but have no formal role in the application process. The companies must apply for the test plates themselves, the city will not be an applicant. HSL (Helsinki Region Transport) has a supporting role in FABULOS. The requests and contacting HSL should go via Ulla Tikkanen.*
- *Traficom is responsible for issuing the test plates. For the exemption and permission related questions directly contact Traficom: automatedvehicles@traficom.fi. Contact persons there: Eetu Pilli-Sihvola, eetu.pilli-sihvola@traficom.fi and Reijo Jälkö, reijo.jalko@traficom.fi. Please cc Ulla and Oscar, oscar.nissin@metropolia.fi for FABULOS related messages!*

Vehicle class / types of registration plates needed?
 Operation allowed with passenger car driving license?
 Insurance issues? (national business ID needed to obtain suitable insurances?)
 Is remote operating allowed? For more than 1 vehicle?
 Liability: operator? company? software owner?
 Mandatory seat belts for field tests? Standing places allowed?
 Ongoing legislation that could have an effect on the field testing?

- Vehicle class / types of registration plates needed? **Test plates.**
- Operation allowed with passenger car driving license? **Yes, for 8+1 ppl. License based on the vehicle type: passenger car (1+8), bus (9+), truck etc.**
- Insurance needed? **Yes, together with national business ID to obtain suitable insurances!**
- Is remote operating allowed? **The law does not mandate a driver inside of the vehicle.**
- Who is liable in case of an accident? **Operator, in case of a driver's mistake or provider in case of system failure.**
- Important to notice that for the insurances required for the exemptions, a national business ID is needed! More information and how to apply: <https://www.ytj.fi/en/index/businessid.html>

Application timeline estimate - explain steps to take
 Application submission: where, how? In English?
 Application content / field test plan / Safety and security plan (content; done by whom?)

Application steps to take (all the steps can be taken in English)

Application process, short form:

1. Prepare your testing plan
 - a. Focus on **safety and risk assessment**
 - b. Contact Traficom with first draft planautomatedvehicles@traficom.fi, Reijo Jälkö (reijo.jalko@traficom.fi), Eetu Pilli-Sihvola (eetu.pilli-sihvola@traficom.fi)
2. When you get approval of the final test plan from Traficom, submit the online application here: <https://www.traficom.fi/en/services/test-plate-certificate>
 - a. Processing time 2-5 days
3. Receive test plates by (snail) mail

Appendices to the online application:

- Trade register extract from the applicant's country of incorporation (not older than three months)
- Proof of insurance cover for third party liability

Testing requirements

- Vehicle under testing **must have a licensed driver** either inside or outside the vehicle
 - License based on the vehicle type (passenger car (1+8), bus (9+), truck...)
- **Test plates** are valid for a year at a time
 - Can be extended

- Cost 200 €
- **Final test and safety report**
 - Incidents, dangerous situations and deviations from the plan must be reported to Traficom at the end of each year

Testing plan requirements

Focus on safety and risk assessment! Topics that should be addressed in the plan:

- **General description of the planned testing**
 - What? Where? When?
 - How many vehicles of what types?
 - During which calendar period will the tests take place?
 - At what times of the day?
 - Who will be operating the vehicles?
 - More detailed information about the drivers
 - Have license for the type of vehicle to be tested
 - Training of the drivers for these specific tests
- **Technical specifications of the test vehicles**
 - Basic technical specifications
 - Differences/modifications from a type-approved vehicle
- **Information about the test area(s)**
 - Maps of the routes/areas
 - Any changes made to the infrastructure (e.g. new traffic signs, changes to right-of-way)
 - How has the plan been discussed with the city/municipality in question?
- **Traffic safety plan**
 - How is safe operation of the vehicle ensured?
 - Risk assessment: severity vs. likelihood
 - How will the vehicle observe its surroundings?
 - How to react / mitigate identified risks?
 - Data collection / sharing
 - Privacy, especially as relates to data collection and sharing (adhering to GDPR requirements)
 - Cybersecurity measures
 - Are there any limitations / restrictions to operate the vehicle (e.g. weather conditions, specific obstacles, requirements to infrastructure)

Road traffic laws and regulations

- **Use of vehicles in traffic**
 - International Conventions on Road Traffic
 - Geneva Convention (1949)
 - Vienna Convention (1968)
 - Managed by UNECE WP.1
 - Objective: harmonisation of traffic rules and road safety
 - Regulations prepared in the EC
 - Mostly DG MOVE
- **Vehicle (system) manufacturing**
 - Mostly in UNECE WP.29

2.3.2. Pilot route and implementation, Helsinki

- Rationale for this route
- Characteristics (incl technical level of traffic light systems, difficult spots & possible infrastructural changes)
- *“Plan B route” with lower ambition level should be available as back-up)*

Pilot route:

The route is located next to Finland’s second busiest train station and an important traffic hub, next to the Helsinki fair center. The route runs around a large block consisting of offices and housing together with several public services and authorities. The robot bus will ride partly on a separate lane, speeds of up to 30 km/h, including right turns, street side parking and a roundabout. There will be at least four bus stops on the route. On demand will be piloted on a small scale in fixed extensions of the route, including left turns and sharing the lanes with trams, buses and private cars.

- *Basic route distance: 1,2 km*
- *Dedicated lane for 1/3 of the route*
- *Only right turns on the basic route*
- *Speed limits: 40 km/h with a separate lane dedicated only to robot bus and 30km/h when lanes are shared with other traffic*
- *1 roundabout and 3 crossings with signalization, currently we don’t foresee integration to smart traffic lights*
- *On demand extension options to serve better people in the offices of the southern part of Pasila and Vallila region.*
- *The parking options are located in the southern and northern end of the extensions*

- Has been discussed and confirmed with (authorities & stakeholders x, y, z)
- Support with legislative matters incl. required licenses and local permission processes.
- Information on city’s software system APIs (depending on technical maturity of each city)

Measures taken to ensure a successful pilot:

- *Two steering groups within the city of Helsinki were formed to ensure the right personnel, resources and circumstances for the piloting:*
 - *High level steering group: includes directors of different departments in Helsinki, together with two mobile operators directors*
 - *Route planning group, consisting of Helsinki city traffic planners (of the pilot area Pasila) together with Traficom and HSL (Helsinki Region Transport) representatives*

- *The Pasila route was originally proposed by Pasila project leader, and it has been agreed with him that temporary arrangements, such as reserving onsite parking for FABULOS purposes, are possible.*
- *The route was inspected in 2018 by the robot bus expert team from Metropolia University of Applied Sciences, who have been designing the previous pilot routes, deploying and operating the pilots.*

- Storage & charging facilities for buses (Storage facilities for the fleet for off-duty hours, electricity for charging (standard socket, 230 V) and s
- Control room location / facilities, pace for control room close to the pilot zone for 2-3 persons.
- Mitigation measures city can/should take; supplier can/should take, Support of local authority's transport planners for the plan on the pilot area & routes.

Storage & charging facilities

- *The parking facilities are found just next to the proposed route, on the southern extension option*
- *Parking will be indoors, with dedicated places reserved for electric cars → parking fees will be covered by FABULOS*
- *There are both slow and fast charging sockets available together with regular 230V sockets → charging fees will be covered by FABULOS*
- *Light maintenance work can be done on the parking lot*
- *There is a dedicated car washing station operated by an independent entrepreneur (costs involved), as well as the possibility to use taxi car washing station (on the condition that the route is high enough and there are no lower parts or obstacles on the route)*
- *The route to the parking has been preliminary inspected, the lowest height exceeds 3m in all circumstances.*
- *In the unlikely situation of not having a high enough space for the vehicles, here is a plan B location within the near proximity of the pilot route.*

Control room

- *It has been agreed that the space for the control room will be provided by Helsinki Region Traffic Management Centre in their Pasila office*
- *The location is in close proximity of the route*
- *The Traffic Management Centre is composed by the Helsinki city along with some neighbour cities, Finnish Transport Infrastructure Agency and southern Finland's police units*
- *Further negotiations on details are ongoing, but it is known that a background screening needs to be carried out for each staff member*

- Please describe the city's open API's (only the ones that are directly related to mobility/ real time location etc)
- Please describe the city's information on wireless network capabilities related to piloting
- Please describe the city's availability of route-wide remote camera surveillance
- Please describe the city's on the vendors and technologies currently used for traffic lights.

Integration opportunities to city's software system APIs

- *Currently we don't foresee integration to smart traffic lights*
- *The integration with the time tables and route planner will be done manually by HSL*

- HSL will introduce the API for the real time location

Integration into journey planner

- Route and stop information of the robotbus to HSL
 - Route will be added to jore, HSL's public transport route and stop system
- Route number for line
 - HSL will give the pilot a line number similar to other lines operating in the area with additional letter R at the end, For instance in the summer of 2019 in Aurinkolahti robotbus 90R operated alongside feeder line 90
- Timetable information to HSL
 - Duration of the pilot
 - Operation days and times
 - Runtime for the route and possibly estimated passing times for other stops
 - If only the runtime is given, HSL will estimate the passing times according to the total runtime and distance between stops
- Departure times
 - Journey planner notice of the pilot
 - HSL information planner adds the notice, link to the project's website
- Disruptions in operation
 - If there are disruptions in the operation of departures that have been added to journey planner, the departures must always be cancelled through OMM (Operational Monitoring and Management), from where the information will go to journey planner, google maps etc.
 - Access to OMM requested from Essi Kyllönen (only in the Phase 3)
 - One account can be used by the whole operation crew

Real-time location into journey planner

- Real time location in journey planner requires the operator to create high frequency positioning data
 - Digitransit's API documentation shows what the end-product should include
 - <https://digitransit.fi/en/developers/apis/4-realtime-api/vehicle-positions-2/>
- Basic rules for HFP data:
 - Send vehicle positioning data 1/s
 - Start sending data with journey information when at first stop waiting/starting journey
 - When not in journey, can send location without any journey info
 - Ask for Operator number from HSL
 - Only VP messages are mandatory for showing basic location at the moment
 - Discuss with HSL if event messages should be implemented per use case
 - Some fields are not mandatory
 - For example if your vehicle does not track stops during route, better leave next_stop empty

- Ask HSL for mqtt broker address and let know your sending ip
 - Testing is always done in dev broker before production

Other open data

- Helsinki Region Infoshare, open data service: https://hri.fi/en_gb/
- APIs: <https://dev.hel.fi/apis/>
- Helsinki Developers' Portal: <https://dev.hel.fi/>
- Helsinki Open Data on GIS and maps, only in Finnish:
<https://www.hel.fi/helsinki/fi/kartat-ja-liikenne/kartat-ja-paikkatieto/Paikkatiedot+ja+-aineistot/avoimet+paikkatiedot/>
- HSL open data & APIs: <https://www.hsl.fi/en/opendata>
- HSL Digitransit: <https://digitransit.fi/en/developers/>

2.4 Lamia

The Municipality of Lamia is home to 74.720 citizens and covers a geographical area of more than 1 million m² area. Moreover, Lamia as a Municipality supports a wide administrative and business zone of influence which covers neighboring Municipalities, resulting in having a population equivalent of 200.000.

The long-term vision for the Municipality of Lamia is to formulate a best practice example in Greece for smart city transport solutions and smart city applications in general. The Municipality is considered a well-equipped city to operate fleets of autonomous vehicles, with self-driving busses already been tested in the country and in neighboring cities with similar demographics, geomorphology, mobility requirements and transportation systems. In general, and on a Municipal but also National level, the maturity for autonomous bus lines deployments is deemed high in terms of regulation, business and operation models, public opinion and acceptance and technical feasibility.

2.4.1. Permissions and exemptions for pilots, Lamia

Describe legislative matters incl. required licenses and local permission processes.

Ecosystem: overview of main Stakeholders - who decides what?

Who needs to be notified? (PTA / PTO (API's, routeplanners), road safety institute, emergency services?)

Greek Government Gazette, Common Ministry Decision [50308/7695](#) of 13/8/2015, entitled "Terms and Conditions for Circulating a Bus without a Driver"

- Provides definitions
- Describes route specifications
- Documents traffic rules
- Describes explicit traffic control room and remote operator specifications
- Provides technical specifications for vehicle registration
- Provides terms and conditions for vehicle operation

*The current legislative framework includes several limitations. There is a common agreement with the Ministry of Infrastructure and Transport for an update of the legislation, towards FABULOS specifications, **which is yet to be finalized.***

The general ecosystem includes the following stakeholders:

- Municipality of Lamia
- Region of Sterea Ellada
- Local Traffic Police
- Ministry of Infrastructure and Transport

*No explicit provision on the language of the documents to be submitted. In our experience, Greek will accelerate the process. **Until further notice, applications and attachments should be communicated***

to the following Lamia Municipality email address: info@lamia-city.gr , while keeping FABULOS representative Fotis Gogoulos in the loop (fotis_gogoulos@hotmail.com).

Vehicle class / types of registration plates needed?
 Operation allowed with passenger car driving license?
 Insurance issues? (national business ID needed to obtain suitable insurances?)
 Is remote operating allowed? For more than 1 vehicle?
 Liability: operator? company? software owner?
 Mandatory seat belts for field tests? Standing places allowed?
 Ongoing legislation that could have an effect on the field testing?

All national provisions are included in the Ministry Decision:

- *No explicit provisions regarding self-driving vehicles types, so in general it depends on the vehicle: M1 or M2-3*
- *Remote operator is considered liable for the AV driving. This liability can be transferred via a specifically designed contract. Needs attention and further examination*
- *No particular regulation for seat belts or standing places. To default approach is to provide seat belts*

Current shortcomings of the legislation:

- *Requirement for a dedicated traffic lane*
- *25km/h maximum speed*
- *Remote operator in charge for monitoring one vehicle*
- *No obstacle overpassing*
- *No dynamic route changes possible*
- *No V2I provisions*
- *No system cyber resilience provisions*

Most important changes agreed (not yet finalized):

- *No dedicated traffic lanes will be required*
- *Enhanced speed limit to 40 km/h*
- *Remote operator can oversee more than one vehicle at the same time*

Application timeline estimate - explain steps to take
 Application submission: where, how? In English?
 Application content / field test plan / Safety and security plan (content; done by whom?)

5 steps to be undertaken

Step	Who	When	Duration (indicative)
Legislation change	Municipality of Lamia and Ministry of Transport	In the following 2 months	Effective immediately
Vehicle registration	Supplier or Research Institution Regional Administration	Can start today, in order to be ready when legislation changes	No indication (must be ready for Phase 3) 2 weeks
Registration plates issuance for Trial Period	Supplier or Research Institution Municipality of Lamia Regional Administration	After vehicle registration	1 month for circulation study + 1 month for the rest, including the training of the remote operator
Trial period	Supplier or Research Institution	After Trial Period registration plates issuance	2 weeks
Registration plates issuance for Piloting	Supplier or Research Institution Municipality of Lamia Regional Administration	After Trial Period	1 week
Vehicle insurance	Supplier or Research Institution	After Piloting registration plates issuance	1 week
TOTAL			3,5 months from the legislation change

In more detail:

1. Vehicle registration:

- Vehicle license/registration issued from EU member state
- Vehicle functional certificate issued from EU member state proving the vehicle has been successfully tested technically according to the provisions of [2009/40 EC](#) and [2010/48 EU](#)
- Technical report from Supplier or Research Institution in charge of the pilot, which describes the means with which the vehicle complies with specific demands (e.g. braking, remote monitoring), as illustrated in the Ministry Decision
- Technical report by a mechanical engineer
 - supplementary technical memo filled
 - compliance of the vehicle with explicit technical specifications (size, weight, lights position, door dimensions e.t.c.)
 - overall design of the vehicle from all view angles

The above are submitted to the Regional Authority, which subsequently performs further tests regarding the functional behavior of the vehicle.

The result of this first step is the issuance of a good condition certificate for the vehicle.

2. Registration plates issuance for Trial Period:

- *Explicit transport and circulation study which indicates minimum infrastructure requirements on specific routes, by the Municipality*
- *Consent by local Traffic Police*
- *Decision by Municipal board*
- *Documentation of the project's objectives and consortium, clearly stating who is in charge of the pilot*
- *Remote operator requirements:*
 - *Valid driving license (D1, D1E, D, DE).*
 - *Valid Professional Passenger Transfer License*
 - *Proof of education on behalf of the Research Institution which has the special needed knowledge for the circulation of the AV.*
- *Testing scenarios which will be carried out during the trial period*
- *Good condition certificate for the vehicle*

3. Trial Period:

- *Vehicle is tested on the basis of certain use case scenarios according to previous steps*
- *Goal is to validate that the AV behaves just as a traditional vehicle*
- *Results are monitored and validated by the Municipality*

4. Registration plates issuance for piloting:

- *Trial period results report by Supplier or Research Institution in charge of the pilot*
- *Updated explicit transport and circulation study if required by Trial Period findings*
- *Final consent by local Traffic Police*
- *Final decision by Municipal board*

5. Vehicle Insurance

2.4.2. Pilot route and implementation, Lamia

- Rationale for this route
- Characteristics (incl technical level of traffic light systems, difficult spots & possible infrastructural changes)
- “Plan B route” with lower ambition level should be available as back-up)

The rationale behind the route is to connect key points of interest of the city, including:

- Intercity bus station
- Chalkiopouleio indoor and outdoor sports center
- National Exposition
- Ampliani outdoor sport center
- Police headquarters
- One of the biggest churches of the city

A simpler and shorter plan B route is available should the plan A be identified as not feasible.

The key challenges of the route are the following:

- Several pedestrian crossings
- One left turn and one right turn in intersection
- Parked cars on the side of the road, especially on wide parts of the road
- Two lanes usually treated as one
- Aggressive driving
- A busy part of the route (0,7 km) lies next to commercial area, which leads to sudden stops
- The same part includes motor vehicle repairers, gas stations, agriculture and constructions material centers, which means heavy vehicles can be part of the route
- Low technical maturity of infrastructure, especially concerning V2I
- High temperatures (can reach 45 °C)

- Has been discussed and confirmed with (authorities & stakeholders x, y, z)
- Support with legislative matters incl. required licenses and local permission processes.
- Information on city’s software system APIs (depending on technical maturity of each city)

The route has been accepted by the current Municipal Board and effectively communicated to the Region of Sterea Ellada and local Traffic Police.

There is strong political will and determination on behalf of the Municipality to successfully host the autonomous bus line.

The Municipality retains specific provisions for supplementary side projects that are deemed necessary for the completion of the route.

There are not any pilot specific APIs available (e.g. traffic lights, traffic, signs)

- Storage & charging facilities for buses (Storage facilities for the fleet for off-duty hours, electricity for charging (standard socket, 230 V) and s
- Control room location / facilities, pace for control room close to the pilot zone for 2-3 persons.
- Mitigation measures city can/should take; supplier can/should take, Support of local authority's transport planners for the plan on the pilot area & routes.

Storage location and Control room are available next to the route and will be provided by the City.

- Please describe the city's open API's (only the ones that are directly related to mobility/ real time location etc)
- Please describe the city's information on wireless network capabilities related to piloting
- Please describe the city's availability of route-wide remote camera surveillance
- Please describe the city's on the vendors and technologies currently used for traffic lights.

There are not any pilot specific APIs available (e.g. traffic lights, traffic, signs)

The underlying route infrastructure is the following:

- 3G/4G coverage
- 100% geographical coverage with ~100 Mbps WiFi connectivity
- Generic APIs available (e.g. GIS data, bus stops, points of interest)
- SWARCO conventional traffic lights with halogen or LED optics

The city is willing to investigate infrastructural changes in regards to route-wide remote camera surveillance depending on the costs.

2.5 Tallinn

In Estonia, the FABULOS procurement partner (member of the Buyers Group) is the Ministry of Economic Affairs and Communications. The field testing in Phase 3 will take place in the capital of Estonia, Tallinn. Tallinn has about 450,000 inhabitants, size 159,3 km².

Licenses and permits are given by Estonian Road Administration, which is a subordinate institution to the Estonian Ministry of Economic Affairs and Communications (the procuring partner, henceforth MKM). MKM is responsible for all the legislation regarding transport (e.g.) in Estonia and also responsible

2.5.1. Permissions and exemptions for pilots, Tallinn

Describe legislative matters incl. required licenses and local permission processes.

Ecosystem: overview of main Stakeholders - who decides what?

Who needs to be notified? (PTA / PTO (API's, routeplanners), road safety institute, emergency services?)

Contact the Road Administration info@mnt.ee They will test the vehicle in order for it to be registered - which is a must in order to be allowed in traffic.

The land owner decides if they allow a vehicle to make public transit rounds, i.e. the city of Tallinn when on city territory (e.g. the airport parking lot/bus station is on city land) and Ülemiste on its territory (inside their campus).

Vehicle class / types of registration plates needed?

Operation allowed with passenger car driving license?

Insurance issues? (national business ID needed to obtain suitable insurances?)

Is remote operating allowed? For more than 1 vehicle?

Liability: operator? company? software owner?

Mandatory seat belts for field tests? Standing places allowed?

Ongoing legislation that could have an effect on the field testing?

Vehicle needs to be registered according to what it is, e.g. how many passengers it transports. The vehicle will be issued a temporary license plate (price 205 euros).

Operation is allowed when the operator has the relevant driving license, i.e. if up to 8+1 places, B category. A human being designated as responsible is a must - Traffic Act § 2 (41): "driving a power-driven vehicle means [...] any activity of a person [which] influences the driving direction or speed of the power-driven vehicle with the help of control devices [...]". i.e. designated responsible person is responsible for any accident – must be diligent

Remote operation is allowed, but the human person is liable for accidents etc, in practice this will be the operator.

The vehicle needs to be insured. Insurance market in Estonia is free, so the company needs to insure the vehicle themselves from insurance companies. Insurance does not have to be from an Estonian company, but needs to be valid in EU. See more:

<https://www.riigiteataja.ee/en/eli/ee/506012015001/consolide/current>

<https://www.iizi.ee/en/eraklient/liikluskindlustus>

Seat belts mandatory - emergency braking for autonomous buses is too dangerous. Unclear whether standing is allowed, depends on the tests of emergency braking.

No changes envisioned in terms of legislation.

For vehicle to be registered:

A technical check of the vehicle to verify the compliance to the requirements of the Directive 2007/46/EC (i.e. those components that are not to do with automated driving should be safe and according to existing regulation, those that are redundant - e.g. parts about the field of view for the driver if there is no driver – will be disregarded and those that are automated – e.g. steering is done via technology such as lidar, radar etc – will be given an exemption).

Closed area test

- Demonstration of driving in manual mode.
- Demonstration how the driver can take over driving function from automated mode.
 - If the hand-over process from automated mode to manual mode starts with the stopping vehicle, then maximum allowed speed is 15km/h.
- Test to check how vehicle reacts to stationary/moving obstacles.
- Test of vehicle with system failures.

Preparation work with vehicle to operate on the track.

- Final examination of vehicle on the track, where should demonstrate that vehicle operates safely in traffic (i.e. ability to prove its capability to do all the functions that you wish to do on the actual route – if a left turn, then demonstrate a safe left turn etc).

Please follow the “Guidelines on the exemption procedure for the EU approval of automated vehicles”:

<https://ec.europa.eu/docsroom/documents/34802/attachments/1/translations/en/renditions/native>

See traffic act here: <https://www.riigiteataja.ee/en/eli/ee/507012014005/consolide/current>

Application timeline estimate - explain steps to take

Application submission: where, how? In English?

Application content / field test plan / Safety and security plan (content; done by whom?)

Application with all technical information mentioned in the Guideline

(<https://ec.europa.eu/docsroom/documents/34802/attachments/1/translations/en/renditions/native>) to the Road Administration (info@mnt.ee). Everything content wise needs to be done by the applicant.

The more information the better. E.g. any past experience in testing, results of tests, official reports from other countries' if has been allowed to operate already in another country (e.g. if already allowed in Finland, then official records proving that).

Timeline estimate: depends on the thoroughness of the application. If everything is in order then the Road Administration does not need more than 2 weeks.

Practically however:

1. Sending application
2. Review by Road Administration
3. Closed area test in Estonia (if materials from tests in other countries are not satisfactory)
4. Registering the vehicle - vehicle needs to be in Estonia
 - a. Inspection of the vehicle (accordance with the directive and the application)

- b. Preparation of the track by the company - mapping etc to prepare for the examination
- c. Final examination

Review of the documents takes 1-2 weeks, closed area test with inspection 1 day, final examination 1 day. The rest is up to the company.

2.5.2. Pilot route and implementation, Tallinn

- Rationale for this route
- Characteristics (incl technical level of traffic light systems, difficult spots & possible infrastructural changes)
- "Plan B route" with lower ambition level should be available as back-up)

Route is in the Ülemiste city, which is the largest privately owned smart city campus in the Baltics, see more: <https://www.ulemistecity.ee/en/>

It is also situated right next to Tallinn airport and Ülemiste mall (one of the biggest malls in Estonia). While Ülemiste and the airport are connected with the city center by a tramline (opened in 2017, takes approx 20 minutes), their main buildings are not connected with each other. I.e. e-Estonia showroom (<https://e-estonia.com/visit-us/>), airport and the mall don't have a connection. There is a daily demand by employees in the campus area and also tourists travelling between these destinations.

The difficult spots:

- left turn from the mall's parking lot (no traffic light)
- maneuvering in the parking lot of the airport (reserved only for public transport)
- interaction with the tramline. A traffic light is installed there, but not operation as of today. If needed for the route, needs to be confirmed with Tallinn city (Jaagup Ainsalu, Jaagup.Ainsalu@tallinnlv.ee)

No official plan B route. We can discuss amending the route together with the company, Ülemiste (Ursel Velve, <ursel.velve@mainorulemiste.ee>) and Tallinn.

- Has been discussed and confirmed with (authorities & stakeholders x, y, z)
- Support with legislative matters incl. required licenses and local permission processes.
- Information on city's software system APIs (depending on technical maturity of each city)

Route has been confirmed.

Support (guidance) will be provided by the Ministry (johann.peetre@mkm.ee) and Road Administration (jyrgo.vahtra@mmt.ee).

Tallinn city APIs: <https://avaandmed.tallinn.ee/eng/> and any questions regarding practicalities, e.g. interfacion, contact Jaagup Ainsalu - he will direct the question to the relevant person within Tallinn Transport department.

GTFS data (Tallinn/Harju):

<http://transport.tallinn.ee/data/gtfs.zip>

GTFS data (whole Estonia):

<http://www.peatus.ee/gtfs>

Complete Tallinn/Harju stops, routes and timetables data are available from two .txt files:

<http://transport.tallinn.ee/data/stops.txt> (about 130KB size)

<http://transport.tallinn.ee/data/routes.txt> (about 350KB size).

Stops and routes departing from each stop are available also in XML format:

<http://transport.tallinn.ee/data/stops.xml>

Real-time vehicle positions (as displayed on <http://transport.tallinn.ee/#tallinna-linn/map>) are available from .txt file in CSV format:

<http://transport.tallinn.ee/gps.txt>

The fields in gps.txt file are transport type, line number, latitude, longitude, speed, heading, vehicle number.

The gps.txt file is updated every 5 seconds.

Real-time departure times (example: for three stops named Kalev as displayed on <http://m.transport.tallinn.ee/#stop/07401-1,07402-1,07403-1>) are available from address:

<http://m.transport.tallinn.ee/siri-stop-departures.php?stopid=1079,1080,1081>

Example of real-time departures data in Tallinn:

Request:

<http://transport.tallinn.ee/siri-stop-departures.php?stopid=1079,1080,1081&time=1390219197288>

The time parameter is used only to make URL unique for each request and so prevent browser from caching the request results. Otherwise the browser may return cached old results when you ask departures for the same stop later.

You may use any value for time parameter that is unique, e.g. date and time with seconds or UNIX time (seconds from 1970-01-01).

Result:

Transport,RouteNum,ExpectedTimeInSeconds,ScheduleTimeInSeconds

stop, 1079

bus,36,50580,50580,Viru

bus,5,51013,51013,Metsakooli

bus,18,51174,51174,Viru keskus

bus,32,51240,51240,Kopli

bus,36,51240,51240,Viru

bus,5,51733,51733,Metsakooli

bus,36,51960,51960,Viru

...

Note: SirilD values from <http://transport.tallinn.ee/data/stops.txt> should be used to specify stops for that request.

Short description of data format used in stops.txt and routes.txt files

The file <http://transport.tallinn.ee/data/stops.txt> contains the data about all stops.

First row indicates column names

ID;Lat;Lng;Stops;Name;Info;Street;Area;City and version of program.

Note, that we might add additional fields in the future, so use column

names to find position of field you need.

Each line in the file has data fields for one stop: stopID, sirilD, GPS coordinates Latitude and Longitude multiplied by 10000, stop name, ids of closest stops for walking, remark, street, area, fare zone, city.

Field values that are equal to values in previous line are removed to reduce file size. So you have to refill the values when processing the file.

The routes.txt file contains the data about all routes and timetables of each route. First row in the file indicates field names. Then one row has fields for route data, the next row has encoded timetable of that route. E.g. lines 3, 5, 7, ... in the file contain routes timetables in encoded format.

Route field values that are equal to values of previous route are removed to reduce file size.

Decoding route timetable:

„ indicates end of row in the timetable. Values in first row are start times of trips (columns) in minutes, values in second row are dates, values in third row are weekdays, values in all next rows are driving times in minutes to next stop in the trip. Use the page <http://transport.tallinn.ee/decode.html> to test the decoding of values into timetable. Look at the source code of this page. The decoding function is named `explodeTimes(encoded_data)`

- Storage & charging facilities for buses (Storage facilities for the fleet for off-duty hours, electricity for charging (standard socket, 230 V) and s
- Control room location / facilities, place for control room close to the pilot zone for 2-3 persons.
- Mitigation measures city can/should take; supplier can/should take, Support of local authority's transport planners for the plan on the pilot area & routes.

Will be provided by Ülemiste.

Storage likely in an old industrial building near the route (as parking house parameters - height of entrance - is wrong).

- Please describe the city's open API's (only the ones that are directly related to mobility/ real time location etc)
- Please describe the city's information on wireless network capabilities related to piloting
- Please describe the city's availability of route-wide remote camera surveillance
- Please describe the city's on the vendors and technologies currently used for traffic lights.

No cameras by the city.

Traffic lights operated by <http://signaal.ee/index.php/et/kontakt>

2.6 Porto

*The Public Transport Operator of the city of Porto, STCP, is one of the Procuring Partners of FABULOS. Please note however, that **Porto will not be a pilot location in Phase 3**. The Portuguese legislation does not allow for testing automated vehicles in mixed traffic. Even though the Portuguese Road Authorities are in the process of adjusting the legislation to allow this, these adjustments will come too late to enable field testing in FABULOS in 2020.*

Instead, in the case that there will be 3 Consortia awarded with Phase 3 contracts, Lead Procurer Helsinki will have 2 pilots (one in spring and one in autumn). Other pilot locations remain the municipalities of Gjesdal, Tallinn, Helmond and Lamia.



Pre-commercial Procurement (PCP) FABU LOS

Future Autonomous Bus Urban Level Operation Systems

Step-by-step field test planning

(part of Tender Document 9: Field Test Specifications)



Step-by-step Field Test planning

The overview below is a more elaborate version of the key steps per stage. This is a tool to help consortia and cities with the planning of their Field Tests; it will also be used for future reporting on Function 9 (Tender Document 2", Functional Specifications; Function 9: Deployment, set-up and service) and some other Functions.

A. Obtaining permissions

As per earlier guidance and city webinars, Suppliers are to obtain all necessary permissions, insurances and test-plates as soon as they are notified of their two pilot locations (relates to all subfunctions of **Non-functional Requirement 3**).

B. Pilot preparation phase

The following step-by-step guidance starts from the assumption that there is permission to operate an automated vehicle in road traffic. That is to say, the necessary test plates and related insurance have been acquired as well as any other relevant permissions.

1. Planning the route

- To be done in cooperation with your site-specific local contact person from FABULOS
- Planning the route (the route itself, bus stops, storage and charging place, etc.).
- Presenting the route to the bus supplier (route map, pictures, videos etc.).
- Supplier visit on site, if needed.
- Supplier makes site analysis of the route where the requirements that have to be materialized are listed in order for the route to be implemented (these can be for instance additional infrastructure or traffic signs to intersections).
- Finding a place for RTK GNSS-base, if needed (usually a roof of the highest building in the area). Recommended option is to use a local virtual reference station (VRS) subscription provider. No need for a fixed RTK base as the satellite corrections come through mobile network. This technique has been used in pilots in Helsinki in the past.

2. Discussing the route with the city or landowner

- For instance, discuss the possibility to remove parking spaces for autonomous shuttle bus stops and making additional arrangements.
- To be done in cooperation with your site-specific local contact person from FABULOS

3. Discussing the route with local PT authority

- To be done in cooperation with your site-specific local contact person from FABULOS
- If there are other PT vehicles driving on the same path, PT authorities usually hope that the robot bus(es) does not disturb the traffic.
- Planning the schedule of the robot bus for synchronising it with other PT vehicles (schedule of the robot bus cannot necessarily be known before the route has been programmed and tested as there are many things that affect the time it takes to drive the route through).

5. Informing local transport safety agency

- To be done in cooperation with your site-specific local contact person from FABULOS
- Only in case this institution was not yet involved in the permission process.

6. Planning all the necessary traffic arrangements for the route

- Traffic arrangements plan for the traffic signs and additional infrastructure. A local traffic arrangement company can be consulted on this.
- Agreement with the city or landowner
- To be done in cooperation with your site-specific local contact person from FABULOS

7. Making all the necessary traffic arrangements for the route

- Traffic signs and other required infrastructure. Your site-specific local contact person from FABULOS can help and a local traffic arrangement company can also be consulted on this.
- Specific bus stop signs for the robot buses or leaflets to normal PT bus stops, if these are used as bus stops (schedule can be difficult to know before the actual operation)
- Make sure it is clear if/how traffic lights communicate with the buses.

8. Making sure that the storage/charging place is ready for the buses

- Site-specific local contact person from FABULOS will bring consortia into contact with the owner/contact person of the storage/charging location.
- Height of space: make sure buses fit.
- Check that there is sufficient space for 3 buses at the same time.
- Safe and secure: make sure it is clear how to gain access and who has access.
- Check warm / cold water for cleaning
- Check that there are suitable charging facilities

9. Acquire unlimited data subscriptions (SIM cards) for the bus and RTK-base

- The responsibility of the Suppliers, but assistance may be needed in foreign countries. In that case, be in touch with the local FABULOS contact person for that particular site.
- In Finland, it was Metropolia University of Applied Science that has acquired these in previous pilots.

10. Transport of vehicles to pilot site

- Obtain transportation quotations if needed
- Take potentially long delivery times into account
- Someone to receive the vehicles and show storage/charging place and the route.
- The Lead Procuring Partner (FV-Helsinki) has a small budget to cover (some of) the transportation costs of the vehicles to the pilot site. To discuss this before paying a transportation company.

11. Fixing the RTK-base

- In case of a RTK-base, the sSupplier needs access to the building's roof and must have permission to install the equipment. Further access during the pilot may be needed as well.
- If VRS technique is used, there has to be a subscription available to upload VRS data in the bus. These are probably also tested beforehand together with the bus supplier and VRS subscription provider.

12. Mapping the route

- Mapping the route with the bus itself or a different type of system.
- If there is busy traffic on the route, the mapping may need to be done during nighttime, as other vehicles may disturb the map quality.
- Mapping can usually be done during one day.

13. Programming of the route

- Route is driven through several times and points such as bus stops, intersections and zebra crossings are programmed.
- Bus stops are to be agreed upon with your site-specific local contact person from FABULOS
- Speed is increased step by step.
- Programming can take several days or weeks depending on the route complexity.

14. Setting up control room

- Space is provided by the FABULOS Procuring Partners (desks, chairs, electricity, wifi, toilets) near the pilot route. For changes or wishes, contact the site-specific local contact person from FABULOS.
- Suppliers are allowed to *not* make use of this provided control room and use one of their own. The control room can be located by the operational area but it is allowed to locate it off-site as well.
- Install all necessary equipment (screens etc).
- If applicable, make sure there is space for a local incident team - in the control room or elsewhere

15. Integration with local PT route planners

- In Helsinki this has been done by informing Helsinki Region Transport (HSL, the public transport operator) about the route and informing about the schedule. HSL has then uploaded the timetables to the local Public Transport Journey Planner.
- To be done in cooperation with your site-specific local contact person from FABULOS, who will bring Suppliers into contact with relevant local Public Transport Operator staff.
- The open HFP API can be used to subscribe to vehicle movements in soft real time so that the buses can be seen in real time in a Journey Planner (<https://digitransit.fi/en/developers/apis/4-realtime-api/vehicle-positions/>). .

C. Implementation phase

1. Monitoring

- Passengers (establish way of counting them)
- Mileage
- Operational hours driven (hrs.)
- Amount of automated and manual driving (%)
- Energy consumption (during the drive, charging energy, specify the consumption of air conditioning/heating?)
- Amount of operator intervention and the situations specified
- Amount of emergency stops
- Realization of the planned timetable (departures, days, hours). What is considered an undriven departure?
 - Example: According to Helsinki's local Public Transport Operator, a departure is considered undriven when a departure has not started before the next departure operating on the same direction, or within 15-20 min of the planned departure.
- Use of the on-demand feature (if available)
- Average speed

2. Reporting

- Report on the points mentioned in C.1 above as well as on general pilot progress and technical and other challenges (e.g. GPS challenges, interference or blind spots) - in the least during mid-term pilot evaluations and end-of-pilot evaluations (with reference to the Deliverables

described in Tender Document 7). Templates will be provided by the Procuring Partners in due time.

- We foresee weekly “update” emails between the Supplier and the site-specific local contact person from FABULOS during the preparation stage and during the actual piloting
- We foresee bi-weekly “update” telephone calls between the Supplier and the site-specific local contact person from FABULOS during the preparation stage and during the actual piloting

3. Communication activities

- In cooperation with the FABULOS site-specific local contact person.
- Take guidelines from the FABULOS Communications team into account.
- Agree on division of tasks between Consortium members and FABULOS team (e.g. related to media contacts; “marketing” videos of pilots; presentations in conferences, materials) and planning.
- To report all communication activities to the site-specific local contact person and the FABULOS Communications team (template will be provided).

4. Customer surveys

- Part of Non-Functional Requirement 2.1 (TD2’). The Supplier is expected to carry out at least one user satisfaction / acceptance survey among a predetermined minimum number of passengers during the Field Testing. Procuring Partners will carry out before and after surveys in their respective cities.
- All surveys will consist of several mandatory questions (equal for all pilot sites) plus the option to add own questions.
- Engage disabled people as part of the survey group.
- Determine who will do the surveys and how. E.g. a person inside the bus or an online link shared for the passengers, QR code, something else?
- Seek cooperation with local/national traveller organisations and/or organisations for disabled people (with reference to NFR2.5) to e.g. determine suitable survey questions, engage/inform their members, invite representatives for test-drives and in general get feedback on your solution.
- Seek cooperation with a University of other stakeholder (with reference to NFR2.5) to analyse and report on the survey results

5. Crisis communications

- Refers to NFR1.10 related to Crisis Communication Plan
- To be done in cooperation with your site-specific local contact person from FABULOS
- Send all relevant actors the agreed-upon procedure in case of an accident.
- Inform all people involved in the crisis communication chain about the exact start date and route of the pilot.

6. Deviations in the operating timetable

- In case of a fixed time schedule: deviations are seen via the local PT journey planner
 - Example: Helsinki’s public transport operator HSL has (during previous pilots) not been very happy with receiving multiple deviation notifications. In case of longer technical failures, the routes have been completely removed from their systems and reloaded again when it is certain that the buses can operate again.
- Consider putting up an own channel for the pilot, where deviations can be reported (example: <https://www.helsinkiroboline.fi/poikkeusinfo/>) . To be done in cooperation with your site-specific local contact person from FABULOS
- Determine who will make the deviation notifications.

- Example: in HSL traffic the operator companies have access to an Operational Monitoring and Management system (OMM) which is used through a browser. These deviations are then automatically transferred to the Journey Planner as the deviation has been registered through OMM.

7. On-demand

- Refers to function 1.9 (TD2”), which is a nice-to-have function: “Routing on the fly to go beyond fixed routes (within pre-specified area of operation), according to passenger on-demand requests”.
- Also refers to function 8.10 (TD2”), which is a nice-to-have function: “Can be operated on fixed routes, stopping anywhere on the route by on-demand request from inside or outside the bus (e.g. mobile app) with safety considered”
- The operational area of the on-demand service needs to be mapped out and taken into account well beforehand in the deployment planning
- Check if the on-demand feature can be integrated in the local public transport route planner
- Communicate the availability of this option to (potential) users
- Bullets above to be done in cooperation with your site-specific local contact person from FABULOS